EC325 Workbook

EC325: Econometrics | Colby College

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2025-12-12

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Preface

About these Notes

This booklet contains the lecture notes for EC 325: Econometrics at Colby College. These notes are meant to supplement the course readings.

The content in this book is *heavily inspired* by three main texts, which are required for this course:

1. Wooldridge (2019)
2. Angrist and Pischke (2014)
3. Wickham, Çetinkaya-Rundel, and Grolemund (2023)
4. Bailey (2020)

Structure of These Notes

This course is basically broken into three modules:

1. Correlation, Causation, and Regression
2. Real-Life Regression Models
3. Introduction to Causal Methods

Each chapter contains both notes for the course - in narrative form - as well as econometric labs that we will work through in R.

Roadmap

- In Unit 1, we will discuss the difference between correlation and causality, and what it means for us as economists.
- In Unit 2 we will learn about the "gold standard" of causal inference–Randomized Control Trials (RCTS). We will also be introduced to the Potential Outcomes Framework, a mathematical formalization of distinguishing correlation from causality.

Statistical Software

Almost all modern econometrics is done with specialized statistical software which you interact with via code.

About Me

I am a recent PhD from UMass Amherst. My research focuses on the relationship between public/economic policy and maternal, infant, and child health and racial/ethnic health disparities. You can find out more about me and my work on my website: www.raymondcaraher.com

1 Introduction

Key Questions

- What is the difference between correlation and causation?
- What does an econometrics research project entail?
- What are the main types of data formats we will work with?
- What is statistical software and why is it useful?

Suggested Readings

- Wooldridge (2019), Ch. 1
- Wickham, Çetinkaya-Rundel, and Grolemund (2023), Ch. 1

1.1 What is this Class About?

This class is obviously dedicated to the study of econometrics (endearingly shortened 'metrics). But what exactly does econometrics entail? What differentiates it from broader terms such as "statistical analysis" or "data analysis"?

From Wooldridge (2019), we get a classic definition of econometrics:

> "…statistical methods for estimating economic relationships, testing economic theories, and evaluating and implementing government and business policy."

However, my view of econometrics is at the same time, broader and more narrow:

Definition of Econometrics

The practice of using statistical, quantitative methods to study social phenomena and provide insights into public policy. More specifically, modern econometrics has focused on identifying the **causal effects** of social phenomena/policy as opposed to **correlations**.

In other words, I view the practice of econometrics as the use of quantitative methods to understand the social world around us, with a special (though not exclusive) lens focusing on separating correlation from causation.

1.2 Does Having Health Insurance *Cause* You to be Healthier?

Let's think about a critical question in public policy: *does having health insurance make someone healthier*? This is especially an important question in the context of the United States, which does not have a public health insurance system. Instead, it is a predominantly private system entangled with a complex net of social safety nets meant to provide a patchwork of coverage to the most marginalized.

However, as of 2023, 9.5% of U.S. adults do not have any health insurance coverage.[1] Would the US as a whole be better off guarantee health insurance coverage for everyone? If health insurance makes people healthier, then there may be a strong case to do so. If not, the benefits of such a system would become, at the very least, more opaque.

Let's take a look at life expectancy in the US (a course but important measure of population health), relative to similar high-income, developed countries. Figure 1.1 shows the average life expectancy from 1990 to 2021 for the US, UK, Canada, Australia, and a number of other high-income countries. As can be seen, the US performs considerably worse in terms of life expectancy, and this divergence between peer countries has gotten *larger* since the 1990s, and especially during the COVID-19 pandemic.

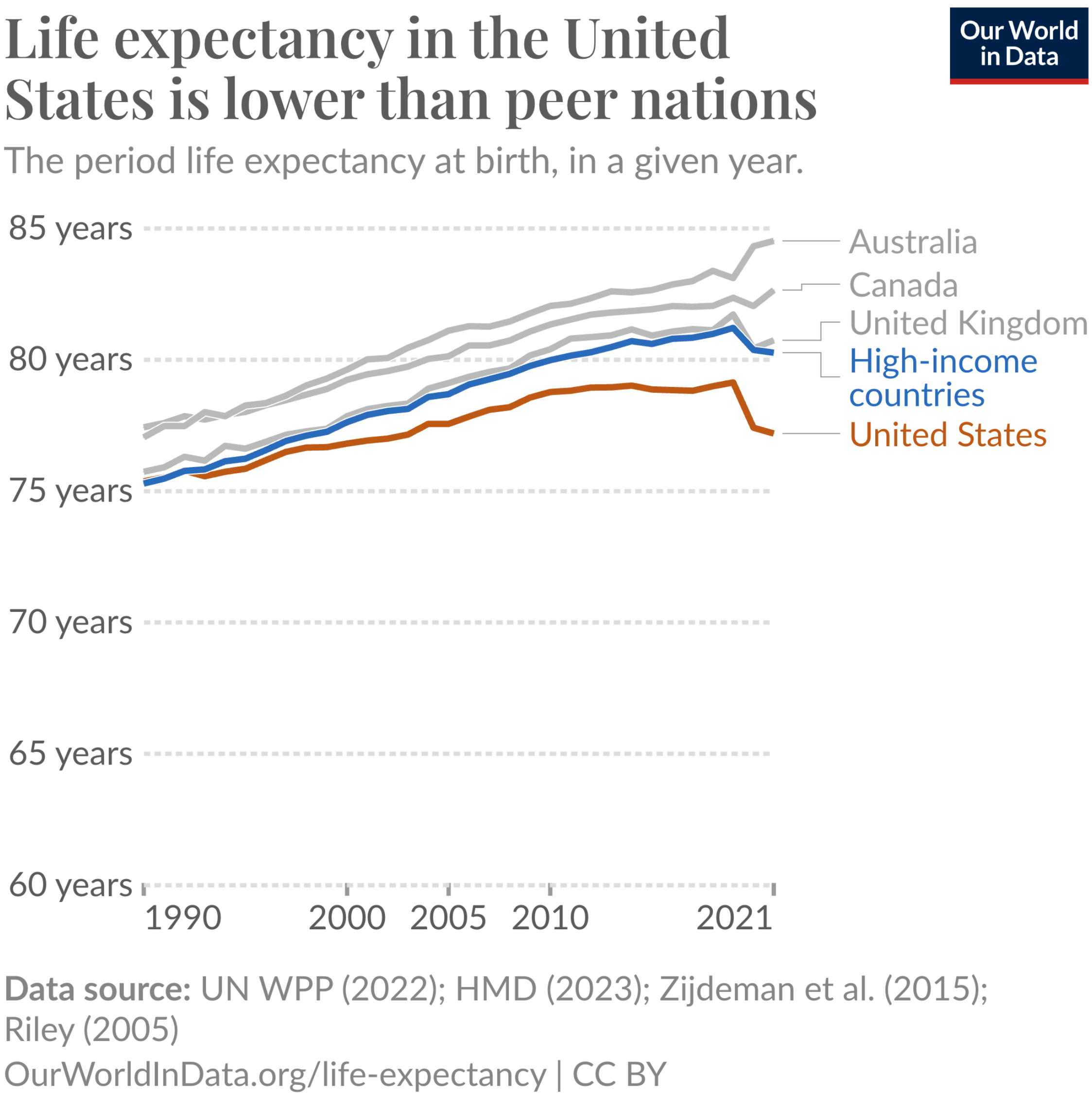
Critically—with the exception of the US—all of the countries in the peer group have some form of universal health care. In other words, all developed countries which are healthier than the US have policies which guarantee the uninsured rate is zero.

This then leads to a critical question: did universal health insurance *cause* these countries to have better health outcomes? Or is this a correlation between policy and outcomes? Certainly, one could make an argument that having health insurance does cause better health outcomes. On the other hand, there are lots of differences between the US and other high-income countries which may also explain the pattern seen in Figure 1.1. For example, maybe these countries have different regulations regarding the food quality, which may explain part of the difference in life expectancy. Or maybe because relatively more people commute to work via bicycle or walking, which may have health benefits compared to driving to work.

These other differences between groups which may *confound* our comparison between countries with universal health insurance and those without also extends to individuals. In the US at least, those with health insurance may be more likely to have a high-paying job which includes insurance as a benefit. Therefore, a comparison between those who have insurance and those who do not will be plagued with the fact that those who have insurance *also are more likely to have a high income job*, who may have had better health outcomes regardless of if they had insurance or not. In other words, we may be confusing *correlation* for **causation*.

[1] Source: KFF

Figure 1.1: Life expectancy in the US vs. peer countries.



Source: Our World in Data

Correlation vs. Causation

The **correlation** between two variables describes how they move together. It means two things are *associated*. **Causation** describes a direct relationship where a change in one variable causes a change in another. It means one thing *makes the other happen.*

A key part of the content in this course is to think about when our comparisons are describing a **correlative** relationship or a **causal** relationship. We will study techniques which—if one is convinced that the requisite assumptions hold—will actually be able to recover the *causal effect* of social phenomena, and bring important insights into essential public policy questions like: what is the effect of health insurance on health outcomes? Does going to college cause you to have higher wages? Do immigrants reduce native wages? and does policing cause decreases in crime?

1.3 Broad Steps of Econometric Analysis

Let's say we want to know what the effect of on-the-job training has on your wage. What steps would we need to take to answer this question?

1.3.1 Step 1: Write the Model

Our first step would be to write an *econometric model,* with the goal of using data and econometric tools to *estimate the parameters of the model.* This model can sometimes be derived from a formal model in economics (i.e., the profit maximization of firms), but it is at least based on some intuitive logic which mathematically relates some variable (the **dependent variable**) to some **independent variable**(s).

An example of the type of econometric model you may want to run to study the effect of on-the-job training is:

$$wage = \beta_0 + \beta_1(training) + \mu, \tag{1.1}$$

where *wage* is the *outcome, left-hand side (LHS), predicted, or dependent variable* (these are synonyms for the same thing), and *training* is the *explanatory, right-hand side (RHS), predictor, or independent variable* (again, synonyms for the same thing). The β_0 and β_1 variables are the parameters that we want to *estimate the value of* using econometric methods, and they tell us *how* our *independent* variable is related mathematically to our dependent variable.

1.3.2 Step 2: Find and Explore the Data

The next thing you have to do—after we figure out the model we want to estimate, such as in Equation 1.1—is to get the data. There is a *huge* range of datasets out there, and we are especially lucky that the data-gathering infrastructure in the United States is incredibly robust. As we go through the semester, we will work with as many different datasets that we can, which cover a range of topics: from sports to health, crime, and economic development. While different datasets usually can their own quirks, broadly speaking there are three different types of data we will work with in this class:

1. **Cross-sectional data:** This is data which contains information on units (e.g., individuals, countries, firms, households, etc.) at a *single point in time.* This data is the most common, and also is generally the simplest to work with, so we will start out by working mostly with cross-sectional data.

2. **Time-series data:** This data consists of observations on a single unit (e.g., a country's Gross Domestic Product (GDP), a specific company's stock price, or the national unemployment rate) across multiple time periods (e.g., years, quarters, months, or even minutes). In time series data, a crucial characteristic is the temporal ordering of the observations. This structure is often used for forecasting and modeling how a variable's past values influence its present and future values.

3. **Panel (or Longitudinal) data**: Panel data combines both cross-sectional and time-series dimensions. It consists of observations on multiple units (e.g., individuals, firms, or countries) over multiple time periods. It essentially tracks the same set of cross-sectional units over time. This gives it more information, more variability, and allows researchers to use a rich set of methods to really hone in on the causal effects of whatever social phenomena being studied.

In all of these cases, econometric data is organized in what is called a **matrix** or a **dataframe**. It should look roughly similar to an Excel Spreadsheet, where each column represents a different variable (which is usually labelled, sometimes not), and each row represents the values that a specific unit has across these variables.

For example, the data we use to estimate the model in Equation 1.1 may look like Table 1.1,

where the **person_name** column is (obviously) each person's name, **person_id** column corresponds the unique numerical identifier associated with each person (this will be more common to see in datasets since names are usually anonymous), the **wage** column shows each person's hourly wage in dollars, and the **yrs_training** column shows the number of years each person received on-the-job training. This row-column structure of organizing data is so fundamental that the word *column* is often interchanged with the word *variable.*

When you first look at some data, it is really important to figure out what each row represents. In the simple example shown in Table 1.1, this is straightforward since each row represents a different person. However, in more complex datasets, it may not be immediately obvious what

Table 1.1: Relationship between training and wages for 10 workers

person_name	person_id	wage	yrs_training
Paul	1	18.50	1.00
Ringo	2	22.00	3.00
George	3	17.65	2.50
John	4	27.95	8.20
Linda	5	19.20	0.50
Yoko	6	24.80	6.80
Pattie	7	20.50	5.50
Maureen	8	23.75	4.20
Cynthia	9	16.40	0.80
Barbara	10	28.50	9.10

each row represents. For example, in panel data, each row may represent a specific person at a specific point in time. So row 1 may be Paul in year 2020, row 2 may be Paul in year 2021, row 3 may be Ringo in year 2020, and so on. Figuring out what each row represents is *critical* to understanding the data, as well as how to appropriately analyze it.

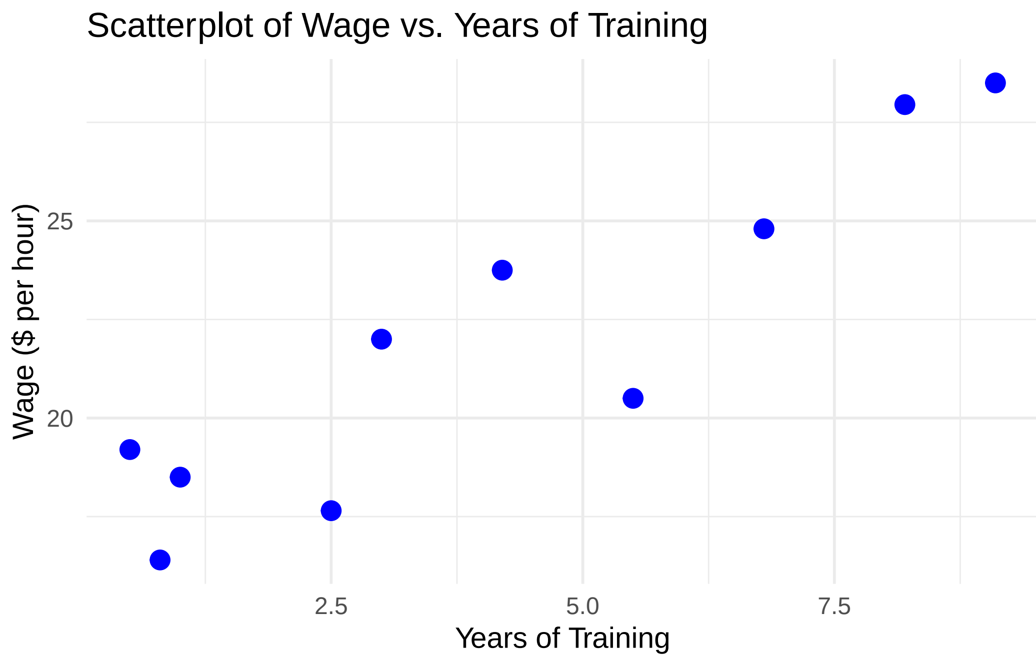
Once we have the data in hand, then we may need to do some data cleaning and organization. This may involve: - Removing missing values - Creating new variables - Changing the format of variables (e.g., from string to numeric) - Merging datasets together - Filtering the data to only include certain observations - And many other tasks.

These are all tasks that in many cases take up the bulk of the time in an econometric research project, and while not the most glamorous part of econometrics, it is absolutely essential to get right.

Before moving on to estimation, it is also important to do some **exploratory data analysis**, (which should be considered step 2.5 in the econometric research process). This involves looking at summary statistics of the data (means, medians, standard deviations, etc.), as well as visualizations (histograms, scatter plots, etc.). Exploratory data analysis is important for understanding the data, as well as for identifying potential issues with the data (e.g., outliers, missing values, etc.).

One of the most important parts of exploratory data analysis is to look at the relationship between the dependent variable and independent variable(s) in your model. One way this can be done is using a **scatterplot**, which plots each observation in the dataset using the independent and dependent variables as coordinates, with the independent variable on the x-axis and the dependent variable on the y-axis.

Figure 1.2: Relationship between training and wages for 10 workers



In Figure 1.2, we see a scatterplot of the relationship between years of training and wages for the 10 workers shown in Table 1.1. From this plot, we can see that there is a positive relationship between years of training and wages: as years of training increases, wages also tend to increase. While not good enough to draw any strong conclusions, one can see how this type of exploratory data analysis is useful for understanding the data and the relationship between variables.

1.3.3 Step 3: Estimate the Model

Once we have a model in mind as well as the appropriate data, we can then use our econometric methods to estimate the relationship. The bulk of this class is dedicated to this step in the econometric research process, but the other stages are no less important! The most common way that econometric models are estimated is with **Ordinary Least Squares** (OLS), which you may remember from your pre-req. In addition to estimating the actually value of the parameters in the model, we also want some idea of how *uncertain* we are about the estimated parameter value. This is where notions of statistical significance and confidence intervals come in.

1.3.4 Step 4: Interpret the Estimate

Once we have an estimated number for the parameter in our model, we then have to interpret the number. Here, the phrase "interpret the estimate" means two parallel things. First, there is the literal, mathematical interpretation of the model. This will change depending on the functional form we use to estimate the model, and we will spend lots of time learning about this. The second part of "interpreting" the model is about what the results mean in the context of your broader research question. Is your estimate big or small? Is it economically important? Do you actually think it reflects a causal effect? These are all types of interpretation which are a critical part of econometrics research, but for which there is not usually a formula we can use. A large part of the "art" working on an econometrics project is exactly this type of interpretation, and how you form it into a compelling narrative. While there are some tools we will cover which will help in this regard, in my opinion, nothing will help you more with this than just reading lots of good, well-written econometrics papers.

1.4 Statistical Software

Much of the econometric process is done using specialized statistical software. We will use this software to: "clean" our data and organize in the appropriate row-column format to do our econometric estimations; explore our data using descriptive statistics (means, medians, etc.) and visualizations (an increasingly important part of a compelling econometric project); estimate our model and conduct analyses of statistical uncertainty; and output our results to easy-to-interpret tables and figures.

There are a few commonly used econometrics software:

- **Stata:** Proprietary software purpose-built for econometrics
- **R:** Open-source programming language with a focus on statistical analysis
- **Python:** Open-source programming language with a focus on data science more broadly
- **EViews:** Proprietary software with a focus on forecasting and other time series applications

Each one of these software of their own pros and cons, and it is worth knowing several quite well. For the purposes of this class, we will learn the **R** programming language to do econometrics. If you haven't take the time to review Appendix A to learn more about R, how to install it, and to run your first chunk of code!

1.5 Summary and Conclusion

This chapter introduced the fundamental framework that will guide our study of econometrics throughout this course. We defined econometrics as the use of quantitative methods to understand social phenomena and inform public policy, with a particular focus on distinguishing between correlation and causation. Through the example of health insurance and health outcomes, we saw how patterns in data can be misleading. While the United States lags behind peer countries in life expectancy, and those peer countries all have universal health insurance, we cannot immediately conclude that universal coverage causes better health outcomes. The tools you will learn in this course are designed to help you identify causal relationships rather than mere associations.

1.5.1 Key Takeaways:

- **Econometrics vs. correlation:** Modern econometrics focuses on identifying causal effects, not just correlations between variables
- **Four steps of econometric research:**
  1. Write an econometric model
  2. Find and explore the appropriate data
  3. Estimate the model parameters
  4. Interpret the results
- **Three main data types:** Cross-sectional (single point in time), time-series (one unit over time), and panel data (multiple units over time)
- **Statistical software:** We will use **R** to clean data, estimate models, and present results

As we move forward, remember that econometrics is as much an art as it is a science. The mathematical tools we will learn are powerful, but they are only as good as the questions we ask and the care with which we interpret our results.

1.6 Check Your Understanding

Note: This section contains interactive content only available in the HTML version.

2 Causality and Randomized Control Trials

Key Questions

- Why can't we interpret correlations as causal effects?
- What is selection bias and how does it arise?
- How do randomized control trials eliminate selection bias?
- What is the potential outcomes framework and how does it formalize causal inference?

Suggested Readings

- Angrist and Pischke (2014), Ch. 1

In Chapter 1, we discussed how econometrics focuses on identifying **causal effects** rather than mere correlations. We used the example of health insurance: while people with health insurance tend to be healthier than people without insurance, we cannot immediately conclude that health insurance *causes* better health. This chapter develops the conceptual and mathematical tools we need to think rigorously about causality, and introduces the randomized control trial (RCT) as the gold standard for causal inference.

2.1 The Problem with Simple Comparisons

Let's return to our health insurance example. People with health insurance are, on average, healthier than people without it. This correlation is undeniable. But can we interpret this as evidence that insurance *causes* better health?

The problem is that people who have health insurance differ from those who don't in many ways beyond just their insurance status. Those with insurance tend to be wealthier, more educated, and have more stable employment. All of these factors independently affect health outcomes. When we compare the health of insured and uninsured individuals, we're not just comparing the effect of insurance—we're comparing people who differ along many dimensions.

Selection Bias

Selection bias occurs when we compare groups that differ systematically in ways beyond just the treatment we're studying. In other words, we are comparing **apples to oranges**.

The comparison we *want* to make is between two otherwise identical people who differ only in whether they have health insurance. In other words, we want an **apples to apples** comparison. This is the essence of the *ceteris paribus* condition: **all else being equal**, the difference between treated and untreated units reveals the causal effect of treatment.

2.2 The Multiverse of Possibilities

To build intuition for causal inference, consider the television show *Rick and Morty*. In this show, the character Rick possesses a "portal gun" that allows him to travel across infinite parallel universes. In each universe, there exists a version of his grandson Morty who is identical in every way except for one characteristic—Cowboy Morty, Evil Morty, Hammer Morty, and so on. Each Morty has the same home, same parents, same sister, same *everything else*.

This multiverse provides the perfect setting for causal inference! Comparisons between different Mortys are truly *ceteris paribus* comparisons because everything except the one characteristic of interest is held constant.

2.2.1 The Causal Effect of College

Suppose we want to know the causal effect of attending college on wages. We can't simply compare the wages of college graduates to non-graduates because these groups likely differ in many unobserved ways (motivation, family background, innate ability, etc.).

But imagine we could observe two versions of Morty from parallel universes: Morty C-137 (who didn't attend college) and College Morty (who did). These two Mortys are identical in every way—same intelligence, same family, same opportunities—except that one went to college and the other didn't.

The difference in wages between College Morty and Morty C-137 would be a genuine *ceteris paribus* difference—the true causal effect of college on wages.

Figure 2.1: Rick and Morty: A show about infinite parallel universes.

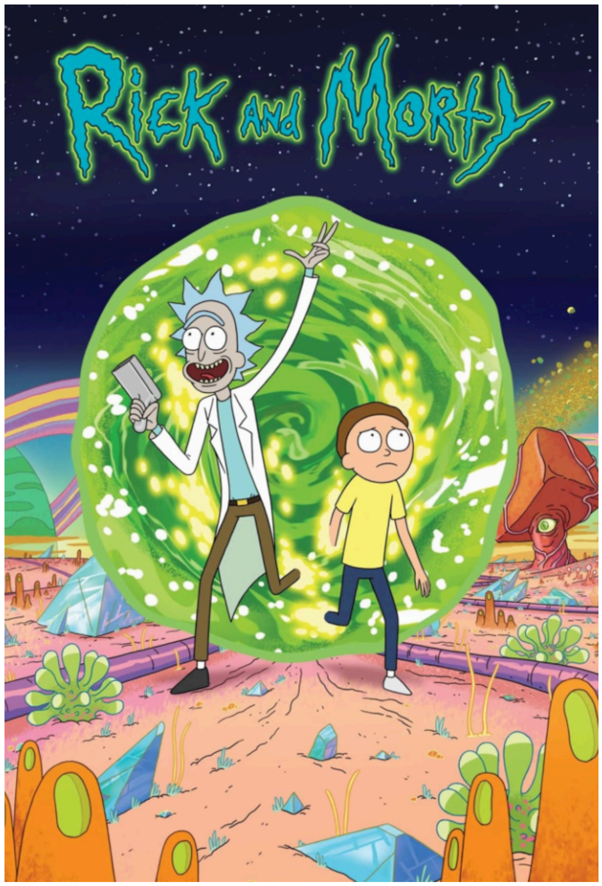
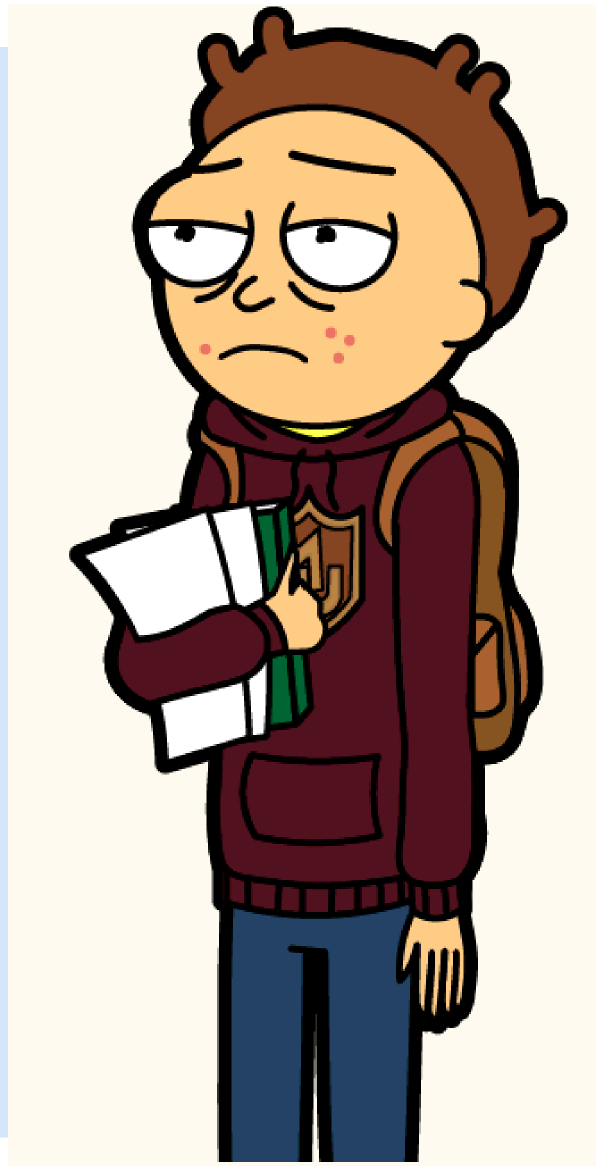


Figure 2.2: Morty C-137 (left) and College Morty (right). The wage difference between them represents the true causal effect of college.

(a) Morty C-137

(b) College Morty



2.2.2 The Fundamental Problem

Of course, the fundamental problem of causal inference is that **we don't have a portal gun**. We only observe one version of reality, not parallel universes. We can never observe what would have happened to a college graduate had they *not* gone to college, or what would have happened to a non-graduate had they attended. This unobserved "what if" scenario is called the **counterfactual**.

2.3 Randomized Control Trials: The Next Best Thing

Since we can't travel between parallel universes, we need another way to make valid causal comparisons. The solution that comes closest to achieving *ceteris paribus* conditions in the real world is the **randomized control trial** (RCT).

Consider how a pharmaceutical company tests whether a new drug works. They don't just give the drug to sick people and see if they get better—that would be subject to all sorts of selection bias. Instead, they conduct an RCT:

1. **Sample** a group of volunteers from the population of interest
2. **Randomly assign** some volunteers to receive the drug (the "treatment" group) and others to receive a placebo (the "control" group)
3. **Compare** the average outcomes between the treatment and control groups

The key insight is that **random assignment eliminates selection bias**. When treatment is assigned randomly, there is no systematic relationship between who receives treatment and their underlying characteristics. On average, the treatment group and control group will be identical in terms of age, health status, income, motivation, and every other characteristic—both observed and unobserved.

> **!** Why Randomization Works
>
> Random assignment ensures that, on average, the treatment and control groups have identical characteristics. Any difference in outcomes between the groups can therefore be attributed to the treatment itself, not to pre-existing differences between the groups.

2.4 The Law of Large Numbers

Why exactly does randomization eliminate selection bias? The answer lies in a fundamental result from probability theory: the **Law of Large Numbers** (LLN).

The LLN states that as the sample size increases, the sample average converges to the population average. Formally, if $Y_1, Y_2, ..., Y_n$ are independent, identically distributed random variables with mean μ, then:

$$\text{plim}(\bar{Y}_n) = \mu$$

where $\bar{Y}_n$ is the sample average and plim denotes the probability limit (what the value converges to as $n \to \infty$).

When we randomly assign individuals to treatment and control groups, we are essentially drawing random samples from the same underlying population. The LLN guarantees that, with a large enough sample, the average characteristics of each group will converge to the population average—and therefore to each other.

2.4.1 A Simulation

> **What is a Simulation?**
>
> A **simulation** is a computer-based experiment where we generate artificial data according to rules we specify. Unlike real-world data, we know exactly how the simulated data was created—including the true causal effects, which variables influence which outcomes, and how treatment was assigned.
> Simulations are powerful learning tools because they let us see what *should* happen under ideal conditions. If we set the true causal effect to zero and then compare treated and untreated groups, any difference we observe must be due to something other than a real treatment effect (like selection bias or random chance). By manipulating features of the simulation—such as sample size or how treatment is assigned—we can build intuition for how these factors affect our ability to recover causal effects. Think of simulations as a laboratory where we can run controlled experiments on statistical methods themselves.

To see the Law of Large Numbers in action, consider a simulation where we have a population with three observable characteristics: light-colored eyes, dark hair, and being tall. Suppose these characteristics are correlated with both treatment status and outcomes in observational data (i.e., without random assignment). Critically, we set the true causal effect of treatment on the outcome to **zero**.

> **A Note on the Code Below**
>
> The R code in the following sections is more advanced than what we've covered so far in the course. Don't worry if you don't understand every line! The code is provided for those who are curious and want to see how simulations work "under the hood." Even if the syntax is unfamiliar, reading through the comments can help you understand the

logic of what we're doing. As you progress through the course, more of this code will start to make sense.

Show Code: Generating the Simulated Data

```r
# Load the tidyverse package for data manipulation and plotting
library(tidyverse)

# Set a "seed" for reproducibility
# This ensures we get the same random numbers each time we run the code
set.seed(181247)

# Define the TRUE causal effect of treatment
# We're setting this to ZERO so we know any observed difference is bias!
causal_effect <- 0

# Create a population of 10,000 individuals
n_population <- 10000

# Generate three random characteristics for each person
# Each characteristic is randomly 0 or 1 (like a coin flip)
eye_light <- sample(x = c(1, 0), size = n_population, replace = TRUE)
hair_dark <- sample(x = c(1, 0), size = n_population, replace = TRUE)
big_feet <- sample(x = c(1, 0), size = n_population, replace = TRUE)

# Generate random "noise" - unexplained variation in outcomes
err <- rnorm(n_population)

# Create an underlying "propensity" for treatment
# Notice: people with light eyes are MUCH more likely to be treated (coefficient = 2.1)
# People with dark hair are moderately more likely (coefficient = 0.83)
# Big feet has a small negative effect (coefficient = -0.11)
treatment_propensity <- 1 + 2.1 * eye_light + 0.83 * hair_dark - 0.11 * big_feet + err

# Assign treatment based on this propensity (treated if propensity > 2.25)
# This creates SELECTION BIAS: treatment is correlated with characteristics
treated <- ifelse(treatment_propensity > 2.25, 1, 0)

# Create the outcome variable
# The outcome depends on the same characteristics that predict treatment!
# But notice: we add "treated * causal_effect" which equals zero
# So treatment has NO real effect on the outcome
outcome <- 1 + 2.2 * eye_light + 1.2 * hair_dark - 0.01 * big_feet + treated * causal_effec

# Combine everything into a data frame
dat <- tibble(
  outcome = outcome,
  treated = treated,
  eye_light = eye_light,
  hair_dark = hair_dark,
  big_feet = big_feet
)

# Let's see how many people ended up in each group
cat("Number in each group:\n")
```

Number in each group:

```r
count(dat, treated)
```

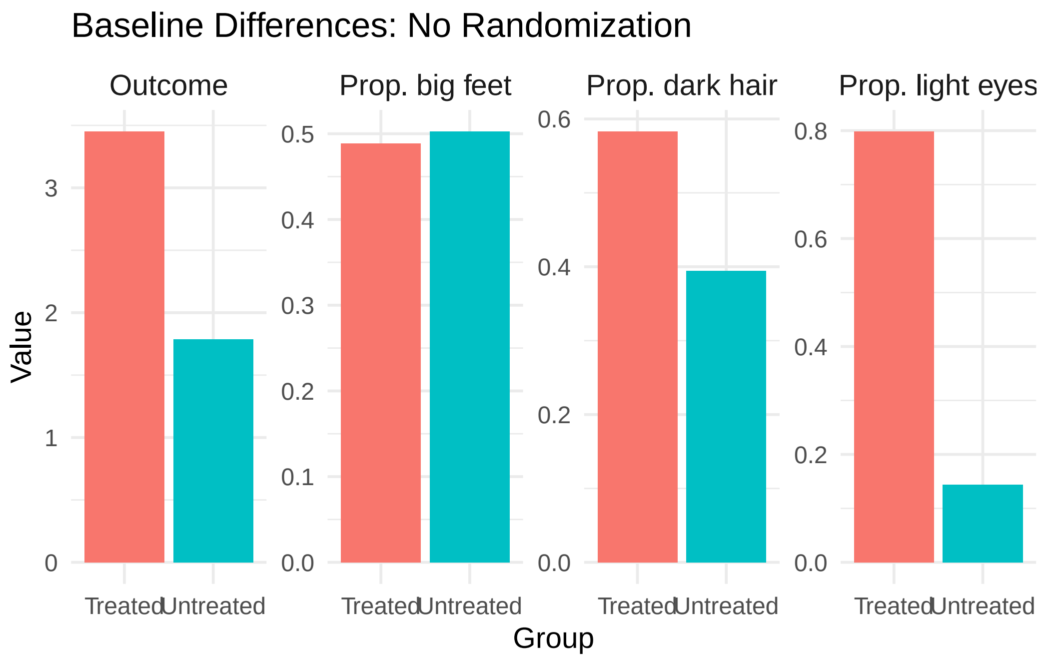
```
# A tibble: 2 x 2
  treated     n
    <dbl> <int>
1       0  4624
2       1  5376
```

When we simply compare treated and untreated individuals *without* random assignment, we see large differences between groups—not just in outcomes, but in their baseline characteristics. The treated group has more people with light eyes, more people with dark hair, and so on. These differences in characteristics lead to a misleading difference in outcomes, even though the true causal effect is zero.

Show Code: Visualizing Baseline Differences (No Randomization)

```r
# Calculate the mean of each variable for treated vs untreated groups
baseline_diffs <- dat |>
  # Group the data by treatment status
  group_by(treated) |>
  # Calculate summary statistics for each group
  summarise(
    Outcome = mean(outcome),
    `Prop. light eyes` = mean(eye_light),
    `Prop. dark hair` = mean(hair_dark),
    `Prop. big feet` = mean(big_feet)
  ) |>
  # Reshape data from "wide" to "long" format for plotting
  pivot_longer(
    cols = Outcome:`Prop. big feet`,
    names_to = "variable",
    values_to = "value"
  ) |>
  # Create nice labels for the treatment groups

  mutate(group = ifelse(treated == 1, "Treated", "Untreated"))

# Create the bar plot
ggplot(baseline_diffs) +
  geom_bar(
    aes(x = group, y = value, fill = group),
    stat = "identity"
  ) +
  # Create separate panels for each variable
  facet_wrap(~variable, nrow = 1, scales = "free_y") +
  labs(
    x = "Group",
    y = "Value",
    title = "Baseline Differences: No Randomization"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    strip.text = element_text(size = 11)
  )
```

Figure 2.3: Baseline differences between treated and untreated groups without randomization. Despite the true causal effect being zero, we observe large differences in both characteristics and outcomes.



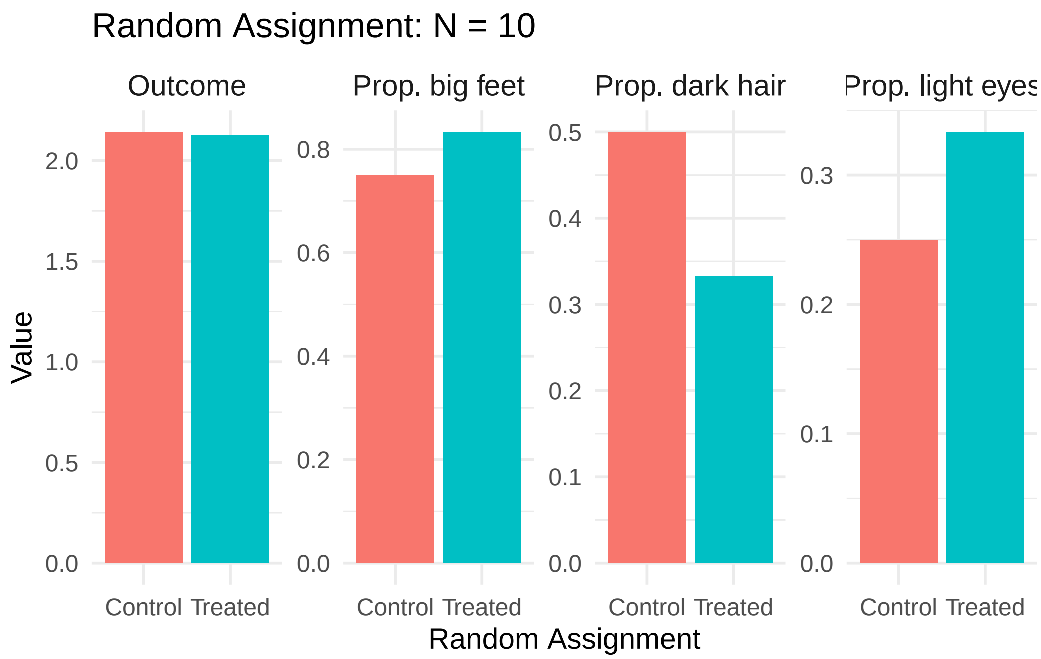
Notice how the treated and untreated groups look very different! The treated group has a higher proportion of people with light eyes and dark hair. Most importantly, the treated group has a higher mean outcome—even though **we know the true causal effect is zero**. This is selection bias in action: people who "select into" treatment have characteristics that also lead to better outcomes.

Now let's see what happens when we randomly assign treatment. Starting with a small sample (N=10), random assignment reduces but doesn't eliminate the differences between groups. With such a small sample, random chance can still produce imbalanced groups.

Show Code: Random Assignment Simulation

```r
# Function to run the random assignment simulation for a given sample size
run_random_assignment <- function(data, sample_size, seed = 181247) {

  # Set seed for reproducibility
  set.seed(seed)

  # Take a random sample from the untreated population
  # (We use untreated only to avoid any prior treatment effects)
  sample_data <- data |>
    filter(treated == 0) |>
    slice_sample(n = sample_size)

  # RANDOMLY assign treatment - this is the key step!
  # Each person has a 50% chance of being assigned to treatment
  # This assignment is INDEPENDENT of their characteristics
  random_treatment <- sample(c(1, 0), size = sample_size, replace = TRUE)

  # Add the random treatment assignment to our sample
  sample_data <- sample_data |>
    mutate(rand_treat = random_treatment)

  # Calculate means for each randomly assigned group
  sample_data |>
    group_by(rand_treat) |>
    summarise(
      Outcome = mean(outcome),
      `Prop. light eyes` = mean(eye_light),
      `Prop. dark hair` = mean(hair_dark),
      `Prop. big feet` = mean(big_feet),
      .groups = "drop"
    ) |>
    mutate(sample_size = sample_size)
}

# Run the simulation for different sample sizes
results_n10 <- run_random_assignment(dat, sample_size = 10)
results_n25 <- run_random_assignment(dat, sample_size = 25)
results_n50 <- run_random_assignment(dat, sample_size = 50)
results_n100 <- run_random_assignment(dat, sample_size = 100)

# Combine all results
all_results <- bind_rows(results_n10, results_n25, results_n50, results_n100)

# Reshape for plotting
all_results_long <- all_results |>
  pivot_longer(
    cols = Outcome:`Prop. big feet`,
    names_to = "variable",
    values_to = "value"
  ) |>
  mutate(
    group = ifelse(rand_treat == 1, "Treated", "Control"),
    sample_size_label = paste("N =", sample_size)
```

Figure 2.4: Random assignment with N=10. Some imbalance remains due to small sample size.



2.5 The Potential Outcomes Framework

Figure 2.5: Random assignment with N=25 (top) and N=50 (bottom). As sample size increases, the groups become more balanced.

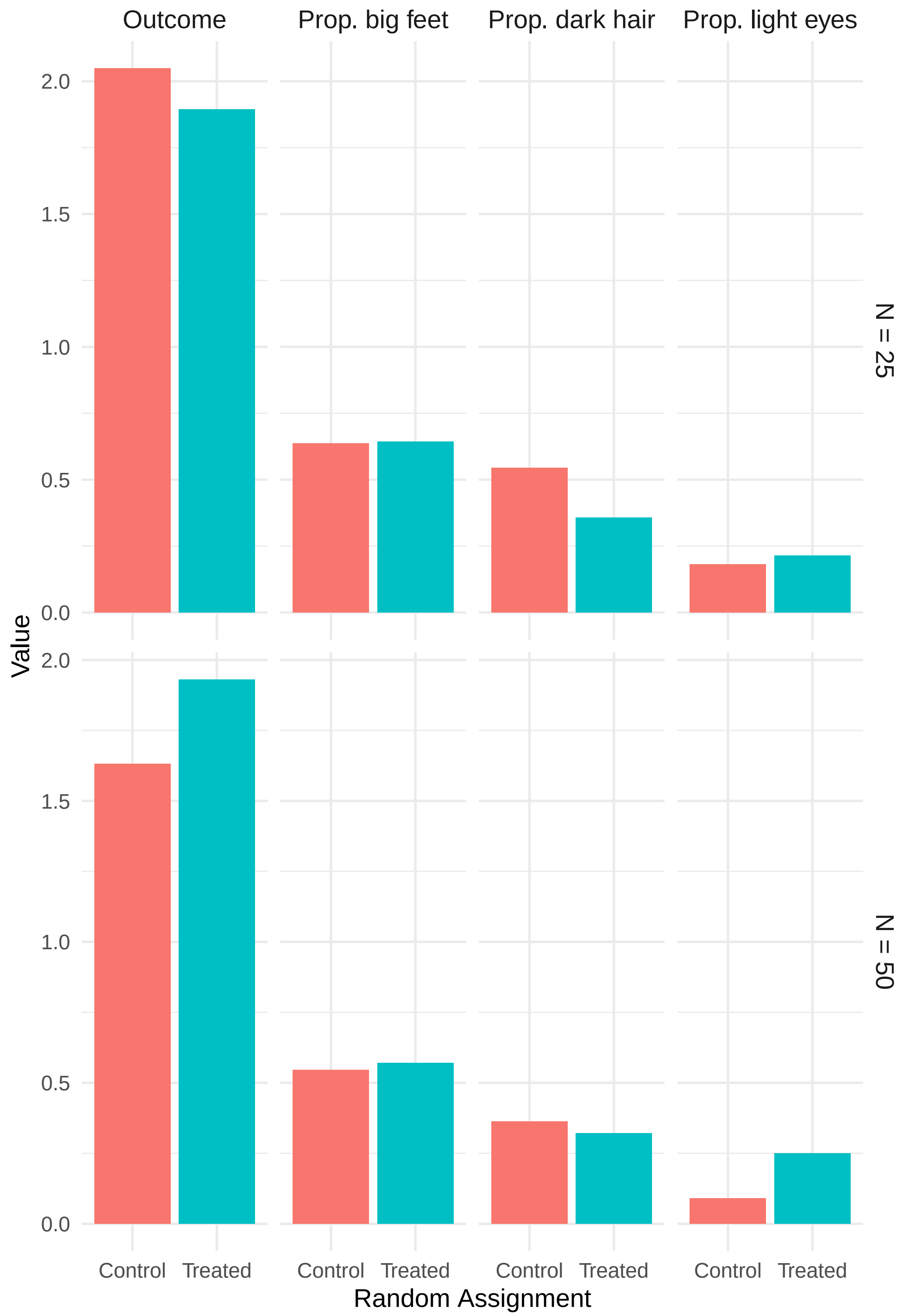
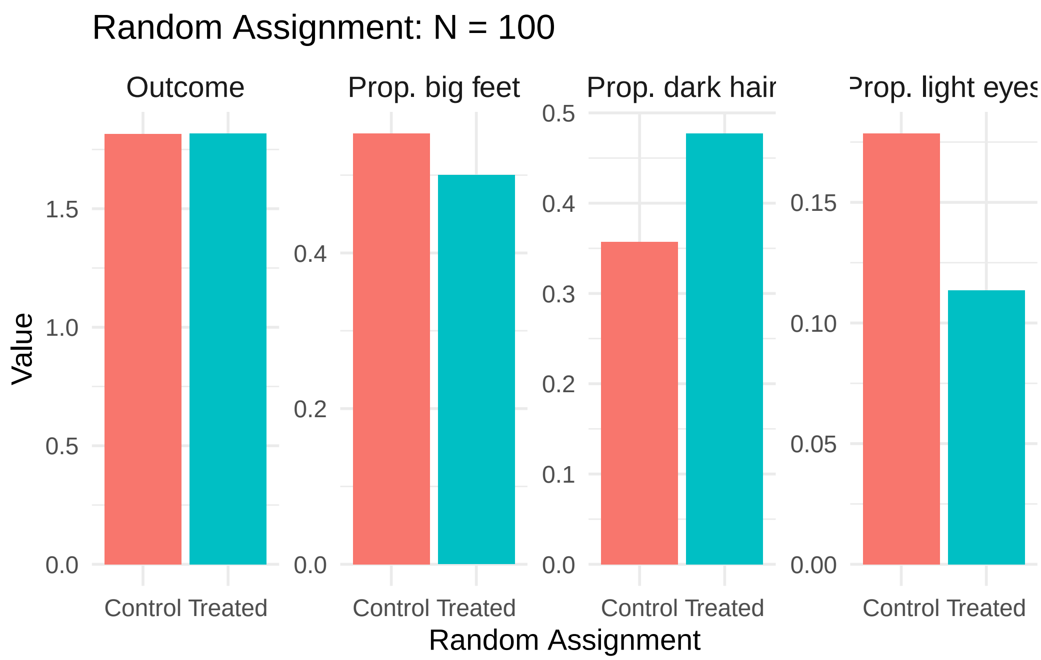


Figure 2.6: Random assignment with N=100. The treatment and control groups are now nearly identical in their characteristics, and the difference in outcomes is close to the true causal effect of zero.



2.5.1 Defining Potential Outcomes

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Figure 2.7: The potential outcomes framework conceptualizes causal effects as the comparison between two paths at a fork in the road.

2.5.2 The Fundamental Problem of Causal Inference

2.5.3 Selection Bias Formalized

❗ The Selection Bias Decomposition

A simple difference in group means equals the causal effect **plus** selection bias:

$$\underbrace{E[Y_i|D_i=1]-E[Y_i|D_i=0]}_{\text{Observed difference}} = \underbrace{\kappa}_{\text{Causal effect}} + \underbrace{\{E[Y_{0i}|D_i=1]-E[Y_{0i}|D_i=0]\}}_{\text{Selection bias}}$$

2.5.4 How Randomization Eliminates Selection Bias

2.6 The Oregon Health Insurance Experiment

2.6.1 Context: Medicaid Expansion

2.6.2 The Lottery

2.6.3 Results

1.
2.
3.

Discussion Question

The Oregon experiment found modest effects of insurance on physical health. Does this mean expanding health insurance is not worthwhile? What other outcomes might matter for policy decisions?

2.7 Summary and Conclusion

2.7.1 Key Takeaways

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2.8 Check Your Understanding

References

A Intro to R and RStudio

Key Questions

- How do I install R and RStudio on my computer?
- What are the main components of the RStudio interface?
- How do I write and run R code?
- What are objects and how do I use them?
- How do I install and load R packages?

Suggested Readings

- Wickham, Çetinkaya-Rundel, and Grolemund (2023), Ch. 1-3

A.1 Installing R

https://cloud.r-project.org/

A.2 Installing RStudio

https://posit.co/download/

rstudio-desktop/

❗ Only Use RStudio

After installing both R and RStudio, you will only ever need to open RStudio. RStudio runs R in the background, so you don't need to open the base R application separately. Feel free to delete the shortcut for base R from your desktop if you wish. When you open an R script file (`.R`) for the first time, be sure to set RStudio as the default program to open such files.

A.3 The RStudio Interface

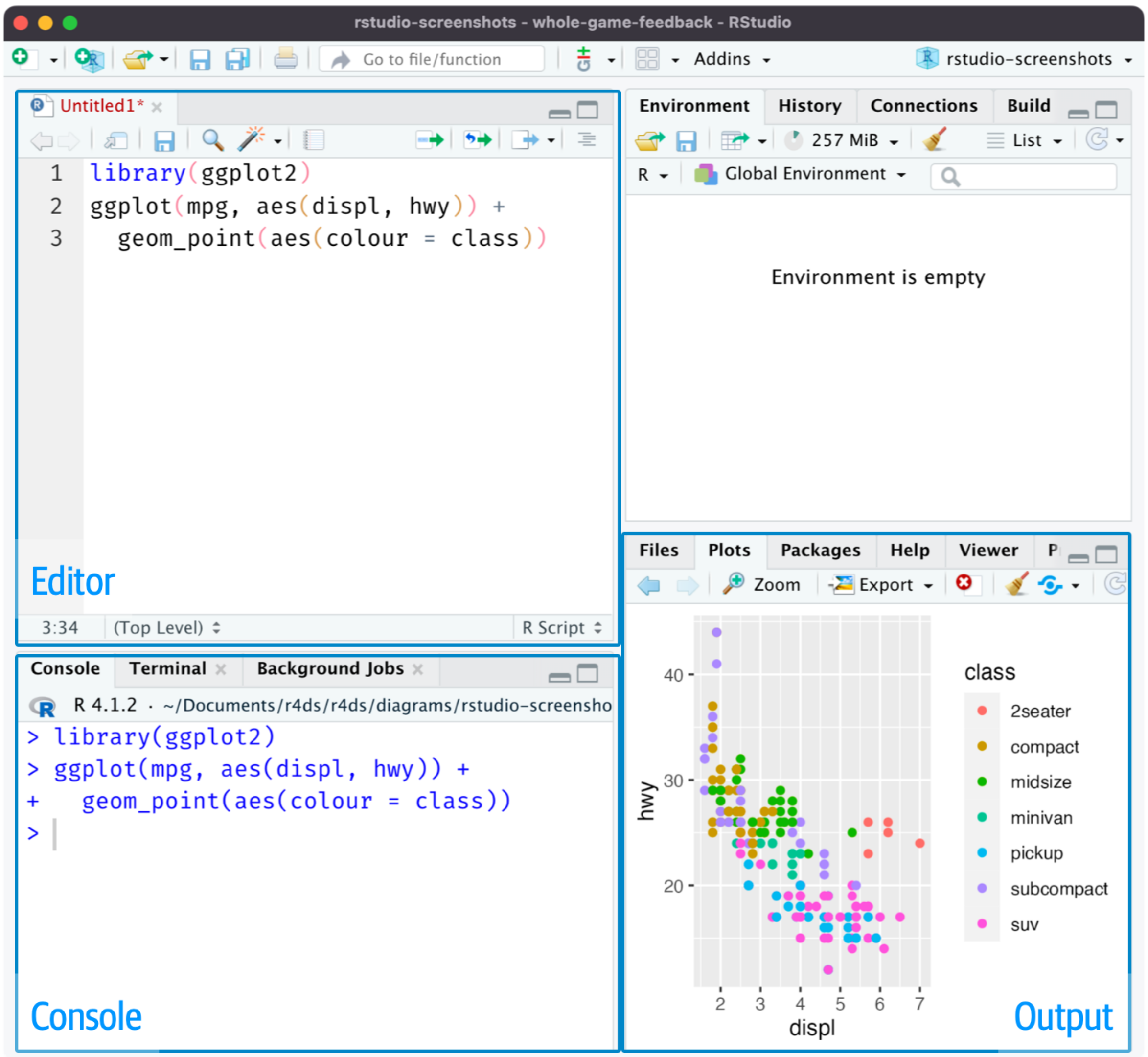
A.1

A.3.1 The Script Editor

A.3.2 The Console

A.3.3 The Environment Panel

Figure A.1: The RStudio interface with its four main panels labeled.



A.3.4 The Plots and Output Panel

Customizing Your Layout

You can hide or show different panels under the "View" menu. This is useful when you want more screen space for a particular task, such as focusing on your script while writing code.

A.4 Basic Coding in R

A.4.1 Arithmetic Operations

```r
2 + 2
```

```r
20 * 5 / 87
```

```r
sin(pi / 2)
```

```r
sqrt(6)
```

A.4.2 Objects and Assignment

```r
x <- 2 + 5
```

```r
x
```

```r
x + 10
```

```r
x <- x + 50
x
```

> Naming Objects
>
> When creating object names, keep them short but descriptive. Object names cannot contain spaces. Some good examples include: `flight_data`, `reg_controls_1`, `acs_male`, or `X_mat`.

A.4.3 Vectors

```
primes <- c(2, 3, 5, 7, 11, 13)
primes
```

```
primes * 2
```

```
primes - 1
```

```
odds <- c(1, 3, 5, 7, 9, 11)
primes + odds
```

A.4.4 Functions

```
bb <- seq(from = 1, to = 30, by = 3)
bb
```

> **Getting Help**
>
> To learn more about any function, type `?` followed by the function name in the console. For example, `?seq` will open the documentation for the `seq()` function.

A.4.5 Data Frames

```r
head(mtcars)
```

```r
nrow(mtcars)
```

```r
ncol(mtcars)
```

```r
colnames(mtcars)
```

```r
head(mtcars$mpg)
```

A.4.6 Comments

```r
# Find the mean miles per gallon
mean(mtcars$mpg)
```

A.5 Working with R Scripts

A.5.1 Creating and Running Scripts

Figure A.2: Example R script with cursor position shown. Pressing Cmd/Ctrl + Enter will run the complete command that creates the `not_cancelled` object, then move the cursor to the next statement.

```
library(dplyr)
library(nycflights13)

not_cancelled <- flights |>
  filter(!is.na(dep_delay), !is.na(arr_delay))

not_cancelled |>
  group_by(year, month, day) |>
  summarize(mean = mean(dep_delay))
```

A.5.2 Script Best Practices

A.6 Working Directories

```
getwd()
```

A.7 R Packages

A.7.1 Installing Packages

```r
install.packages("tidyverse")
```

A.7.2 The Tidyverse

A.7.3 Loading Packages

```r
library(tidyverse)
```

> **!** Remember the Difference
>
> `install.packages()` downloads and installs a package to your computer (do this once). `library()` loads an already-installed package into your current R session (do this every time you start R).

A.8 Summary and Conclusion

A.8.1 Key Takeaways

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A.9 Check Your Understanding

B Statistics and Probability Review

Key Questions

- What are the properties of the summation operator?
- How do we interpret derivatives and partial derivatives?
- How do we find the maximum or minimum of a function?
- What is the difference between a population and a sample?
- What is a random variable and how do we describe its distribution?
- How do we measure central tendency, variability, and relationships between variables?
- What is the normal distribution and why is it important?

Suggested Readings

- Wooldridge (2019), Appendix A, B, and C

B.1 Basic Mathematical Tools

B.1.1 The Summation Operator

B.1.1.1 Properties of Summation

Common Mistakes with Summation

These properties do **not** extend to division or multiplication in the ways you might expect. We **cannot** break up divided expressions:

$$\sum_{i=1}^{n} \frac{x_i}{y_i} \neq \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} y_i}$$

We **cannot** break up multiplicative expressions:

$$\sum_{i=1}^{n} x_i^2 \neq \left(\sum_{i=1}^{n} x_i\right)^2$$

B.1.1.2 The Sample Mean

B.1.1.3 Useful Summation Results

B.1.2 Derivatives

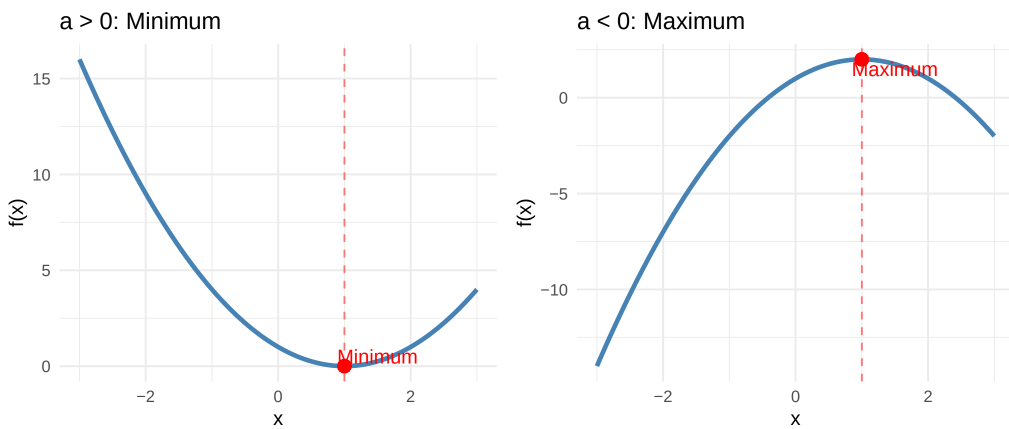
B.1.3 Partial Derivatives

B.1.4 Optimization: Maxima and Minima

B.1.4.1 Finding Critical Points

B.1.4.2 Determining Maximum vs. Minimum

Figure B.1: Left: When a > 0, the parabola opens upward and the critical point is a minimum. Right: When a < 0, the parabola opens downward and the critical point is a maximum.



> **Optimization in OLS**
>
> In OLS regression, we minimize the sum of squared residuals, which is a quadratic function of the coefficients. The second derivative is positive (the function is convex), confirming that the solution is indeed a minimum. This is why setting the first derivative equal to zero gives us the best-fitting line.

B.1.4.3 Optimization with Multiple Variables

B.2 Population vs. Sample

! The Goal of Statistical Inference

The fundamental goal of statistics is to use information from a **sample** to draw conclusions about the **population**. We observe sample statistics (like the sample mean $\bar{x}$) and use them to make inferences about population parameters (like the population mean μ).

B.3 Probability Basics

B.3.1 Random Variables

B.3.2 Probability

B.3.3 Probability Distributions

B.3.4 Expected Values

B.3.4.1 Properties of Expected Values

B.4 Measures of Variability

B.4.1 Variance

B.4.2 Standard Deviation

B.4.3 Covariance

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-
-

B.4.4 Correlation Coefficient

Correlation vs. Causation

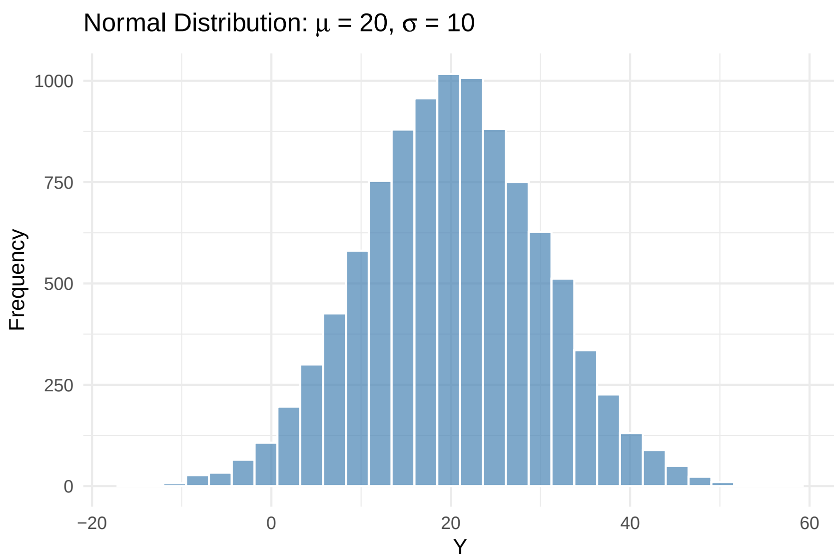
Remember from Chapter 1: correlation does not imply causation! A strong correlation between X and Y tells us the variables move together, but it does not tell us whether X causes Y, Y causes X, or some third factor causes both.

B.4.5 Conditional Expectations

B.5 Distributions

B.5.1 The Normal Distribution

Figure B.2: A histogram of 10,000 draws from a normal distribution with mean 20 and standard deviation 10.



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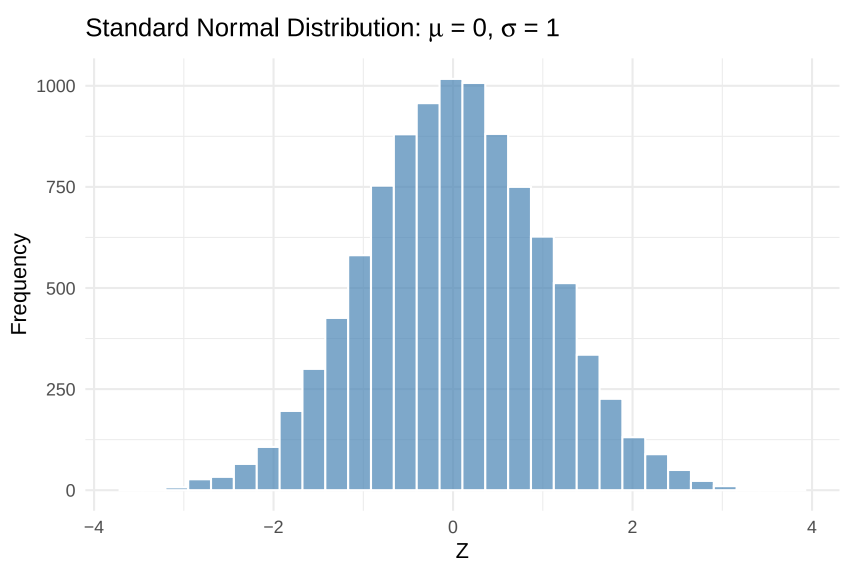
B.5.2 The Standard Normal Distribution

Figure B.3: A histogram of 10,000 draws from a standard normal distribution (mean 0, standard deviation 1).



B.6 Summary

B.6.1 Key Formulas

B.7 Check Your Understanding